

2. **Firestore's Strengths and Limitations:** Firestore's real-time synchronization and authentication services accelerated development considerably. However, complex aggregation queries required client-side processing, highlighting the trade-offs of NoSQL databases for analytics-heavy applications.
3. **Component-Based Architecture:** The modular, component-based approach using React and Radix UI allowed parallel development across team members with minimal merge conflicts. Reusable components (e.g., StatCard, LocationPicker, PhotoUploader) significantly reduced code duplication.
4. **Agile Methodology:** The sprint-based approach allowed for iterative refinement based on testing feedback. Features that were initially planned (e.g., IoT integration) were deferred to future versions based on priority reassessment.
5. **Security-First Design:** Writing Firestore security rules early in the development process ensured that security was not an afterthought. Testing rules with the Firebase Emulator Suite was essential for validating access control logic.

13.3 Social Impact

AgriTrace has the potential to create significant social and environmental impact:

- **Air Quality Improvement:** By diverting agricultural waste from open burning, the platform directly contributes to reducing air pollution in affected regions, potentially improving air quality for millions of people.
- **Farmer Income Generation:** Farmers earn revenue from waste that would otherwise be burned, providing an additional income stream. The platform's transparent pricing ensures fair compensation.
- **Employment Creation:** The collection agent role creates employment opportunities in rural areas, particularly for youth who can leverage smartphone-based platforms for logistics work.
- **Sustainable Development Goals:** AgriTrace contributes to multiple UN Sustainable Development Goals, including SDG 13 (Climate Action), SDG 12 (Responsible Consumption and Production), SDG 3 (Good Health and Well-being), and SDG 8 (Decent Work and Economic Growth).

In conclusion, AgriTrace demonstrates the viability of technology-driven agricultural waste management. With continued development and wider adoption, the platform has the potential to significantly reduce crop residue burning, improve air quality, generate economic value from agricultural waste, and contribute meaningfully to India's sustainability goals.